Teaming Proposal

IT Services

Prepared for: HDR ENGINEERING INC | Prepared by: Ranjeet Kumar — Founder & CEO, OrbitAvanya Tech LLP (AvanyaEdge) | Engagement Reference: OAT-CES-2026-140P6026R0008-FULL | Proposal Date: July 16, 2026 | Validity: 90 days from proposal date

Content

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Executive Summary

OrbitAvanya Tech LLP is pleased to submit this teaming proposal to HDR Engineering, Inc. in response to Solicitation **140P6026R0008** for IT Services. As the prime contractor, HDR brings unparalleled engineering expertise, a **14,200+ employee** global workforce, and a proven **$4.98M SC DOT FY25** track record. OrbitAvanya proposes to contribute a **15% workshare** as a specialized technology subcontractor, delivering our flagship **AI Analytics Dashboard** product to address the data analytics, visualization, and intelligent reporting requirements inherent in modern federal IT service delivery.

Our **AI Analytics Dashboard** is a production-ready, cloud-native platform built on **React, Next.js, Python, FastAPI, Node.js, PostgreSQL, MongoDB, and AWS**. It provides advanced ETL pipelines, real-time analytics, predictive modeling, and interactive dashboards tailored for complex engineering and infrastructure datasets. By integrating our product into HDR's solution, the joint team gains immediate capability in **automated data ingestion, geospatial analytics, asset performance monitoring, and AI-driven decision support** — critical for transportation, water/wastewater, and civil-structural programs.

This proposal outlines a clear subcontract scope, technical alignment matrix, phased delivery plan, milestone-based investment structure, and SLA commitments. OrbitAvanya is prepared to execute a **Team Agreement** and **Subcontract** upon award, with dedicated engineering resources allocated from Day 1. We are confident that our technology contribution will strengthen HDR's competitive position and ensure successful contract performance for the Issuing Agency.

Key Highlights

- **15% workshare** focused on AI/ML analytics, data engineering, and dashboard delivery — complementary to HDR's engineering core
- **AI Analytics Dashboard** product is production-deployed with **Python ETL pipelines, PostgreSQL optimization, and AWS serverless architecture**
- Direct alignment with federal IT modernization mandates: **FITARA, M-19-17, OMB Circular A-130**, and **NIST 800-53** security controls
- **Zero-ramp-up** contribution: product modules are configurable, not custom-built, reducing schedule risk and TCO
- OrbitAvanya holds **AWS Advanced Tier Services Partner** status and **CMMI Level 3** appraisal — ensuring process maturity
- Proposed **Fixed-Price / T&M hybrid** pricing model with **milestone gates** at Sprint 0, MVP, UAT, and Production Cutover
- Dedicated **Product Owner, 2 Full-Stack Engineers, 1 Data Engineer, 1 DevSecOps Engineer** assigned for contract duration
- Past performance includes **analytics platform delivery for a $12M DOT ITS program** and **water utility asset dashboard for 500+ assets**

Respectfully,

Ranjeet Kumar Singh

Founder & CEO
OrbitAvanya Tech LLP (AvanyaEdge)

Strategic Context

HDR Engineering, Inc. has established itself as the premier employee-owned engineering firm serving public sector infrastructure needs across **15 countries and 200+ offices**. The **ESOP model** aligns long-term incentives with client outcomes — a philosophy OrbitAvanya shares as a partner-focused technology firm. Solicitation **140P6026R0008** calls for IT Services that increasingly demand **data-driven decision making, real-time operational visibility, and AI-augmented workflows** across transportation, water, and civil programs.

OrbitAvanya's **AI Analytics Dashboard** was purpose-built for this intersection of engineering and intelligence. Rather than proposing staff augmentation, we offer a **productized capability** that HDR can configure, brand, and deploy across task orders. This subcontracting model preserves HDR's prime accountability while injecting specialized analytics IP that would otherwise require 18–24 months of internal R&D. Our **15% workshare** is scoped to deliver maximum technical leverage per dollar invested — focusing on the analytics layer that amplifies the value of HDR's domain expertise.

Finding 1 · Analytics Gap in Current Engineering Delivery

Federal engineering programs increasingly require **real-time asset dashboards, predictive maintenance models, and cross-program KPI aggregation** — capabilities not native to traditional CAD/GIS/CM toolchains. HDR's current delivery model relies on manual report generation and siloed data stores.

- **68% of DOT program managers** cite "lack of unified analytics" as a top risk (AASHTO 2024 survey)
- Legacy ETL processes average **14-day latency** for project performance data
- No standardized **AI/ML model lifecycle management** across task orders

Finding 2 · OrbitAvanya Product Readiness Assessment

The **AI Analytics Dashboard** has been validated in **3 production environments** serving state DOTs and water utilities. Core modules are containerized, API-first, and **FedRAMP Moderate** aligned.

- **Python ETL Framework**: 200+ pre-built connectors (ESRI, Oracle Primavera, SAP, Maximo, CSV, IoT)
- **PostgreSQL + TimescaleDB** for time-series asset telemetry at **1M+ readings/day**
- **React/Next.js** frontend with **role-based access control (RBAC)** and **Section 508 compliance**
- **FastAPI** microservices exposing **OpenAPI 3.0** contracts for HDR integration
- **AWS Landing Zone** with **Control Tower, Security Hub, GuardDuty — ATO-ready**

Finding 3 · Integration & Compliance Synergies

HDR's **$16.98M annual SC DOT** portfolio and **WTS Employer of the Year** recognition indicate deep transportation domain authority. OrbitAvanya's analytics layer extends this authority into **data-driven program controls** without duplicating engineering workflows.

- NIST 800-53 Rev. 5 control inheritance model: OrbitAvanya provides SC-7, SI-4, AU-6, AC-6 implementation evidence
- FITARA Category 1 reporting automation via dashboard APIs
- OMB M-19-17 cloud-smart alignment: multi-AZ, IaC (Terraform), GitOps (ArgoCD)
- CMMI Level 3 process artifacts (PP, PMC, REQM, VER, VAL) available for DCMA surveillance

Scope of Work

OrbitAvanya's **15% workshare** encompasses the **design, configuration, deployment, and sustainment** of the **AI Analytics Dashboard** as an integrated component of HDR's IT Services solution. Scope includes: (1) **Data Engineering** — building and maintaining ETL/ELT pipelines from HDR's authoritative sources (Primavera P6, ESRI ArcGIS, Oracle, Maximo, IoT sensors); (2) **Analytics & AI** — configuring predictive models for asset deterioration, schedule risk, cost variance, and resource optimization; (3) **Dashboard & Reporting** — delivering role-based portals for program managers, engineers, and executive stakeholders; (4) **DevSecOps & ATO Support** — maintaining the AWS infrastructure, CI/CD pipelines, and security documentation for continuous Authority to Operate; (5) **Knowledge Transfer & Training** — enabling HDR staff to self-service reports, create ad-hoc analyses, and manage model retraining cycles. All deliverables are mapped to **Agile sprints (2-week cadence)** with **Definition of Done** including security scan, accessibility test, and performance benchmark.

Deliverables

- **Sprint 0 (Week 1–2)**: Project Charter, Architecture Decision Records, AWS Landing Zone Deployment, Security Plan (SSP), FedRAMP Control Mapping Matrix
- **MVP Dashboard (Week 3–6)**: Configured **AI Analytics Dashboard** with 5 core data sources, 3 predictive models (schedule risk, cost variance, asset health), 8 role-based views, Section 508 VPAT
- **UAT Release (Week 7–8)**: User Acceptance Test scripts, stakeholder demo recordings, defect remediation (Sev 1/2 = 0 open), Performance Test Report (<2s p95 latency)
- **Production Cutover (Week 9)**: Go-Live Runbook, Rollback Plan, Operations Handoff Package, **ATO Support Package** (POA&M, SAR, Continuous Monitoring Strategy)
- **Sustainment Sprint 1–4 (Week 10–18)**: Model retraining pipeline, 2 additional data source integrations, advanced geospatial analytics module, executive mobile dashboard
- **Knowledge Transfer Package**: Admin Guide, Data Steward Playbook, Model Governance Framework, 4 half-day training workshops (recorded)
- **Monthly Service Reports**: SLA compliance, capacity trends, security posture, enhancement backlog prioritization

Technical Requirements Alignment

RFP Requirement	Our Solution	Alignment
Advanced Data Analytics & ETL for multi-source engineering data	OrbitAvanya **Python ETL Framework** with 200+ connectors; **PostgreSQL/TimescaleDB** for time-series; **Apache Airflow** orchestration on **AWS MWAA**	Full
Real-time dashboarding & visualization for program controls	**React/Next.js** dashboard with **WebSocket** live updates, **RBAC**, **Section 508/WCAG 2.1 AA**, **OpenAPI** for embedding in HDR portals	Full
Predictive analytics / AI for	**Scikit-learn / XGBoost / Prophet** models in	Full

schedule, cost, asset performance	**FastAPI** microservices; **MLflow** tracking; automated retraining via **SageMaker Pipelines**	
FedRAMP Moderate / NIST 800-53 compliance for cloud hosting	**AWS GovCloud (US-East/West)** deployment; **Control Tower** guardrails; **Security Hub** + **GuardDuty** + **Config**; **IaC (Terraform)** with **Checkov** policy-as-code	Full
Integration with HDR enterprise systems (Primavera, ArcGIS, Maximo)	Pre-built **REST/GraphQL** adapters; **OAuth2/OIDC** via **AWS Cognito / Azure AD**; **ETL scheduler** with **incremental CDC**	Full
Geospatial analytics for transportation / water infrastructure	**PostGIS** + **Kepler.gl** integration; **H3 hexagonal indexing**; **GTFS/GTFS-RT** transit feed support	Partial
Mobile-responsive executive reporting	**PWA** with offline sync; **iOS/Android** wrapper via **Capacitor**; **push notifications** for threshold alerts	Partial

Our Proposed Solution

The **AI Analytics Dashboard** is not a custom development effort — it is a **mature, configurable product** that OrbitAvanya will tailor to HDR's task order requirements. At its core, the platform operates as a **three-layer architecture**: (1) **Data Fabric Layer** — **Python-based ETL/ELT pipelines** orchestrated by **Apache Airflow (MWAA)** ingest structured and unstructured data from HDR's systems of record. Connectors for **Oracle Primavera P6 (REST API), ESRI ArcGIS (Feature Service), IBM Maximo (MIF), SAP (OData), and IoT/SCADA (MQTT/Kafka)** are pre-built and maintained. Data lands in a **PostgreSQL/TimescaleDB** warehouse with **dbt** transformations ensuring **dimensional modeling** consistency. (2) **Intelligence Layer** — **FastAPI** microservices expose **ML models** for **schedule risk scoring (XGBoost), cost variance forecasting (Prophet), asset deterioration curves (Weibull/ML hybrid), and resource optimization (OR-Tools)**. Models are versioned in **MLflow**, deployed via **SageMaker Pipelines**, and monitored for **drift (Evidently AI)**. (3) **Experience Layer** — **Next.js/React** frontend delivers **role-based portals**: Program Manager (portfolio KPIs, earned value, risk heatmaps), Engineer (asset drill-down, work order analytics, geospatial map), Executive (mobile-first strategic views, natural language Q&A via **LangChain + Bedrock**). All layers are **containerized (Docker)**, deployed to **AWS ECS Fargate** behind **ALB + WAF**, with **GitOps (ArgoCD)** from **GitHub Enterprise**.

Integration with HDR's environment follows a **zero-trust pattern**: **OIDC federation** with HDR's **Azure AD**, **VPC peering / PrivateLink** for on-prem data sources, **API Gateway** with **mutual TLS** for external consumers. **Security** is baked in: **Secrets Manager** for credentials, **KMS** for encryption, **CloudTrail + GuardDuty** for audit, **Config Rules** for drift detection. **Accessibility** is validated per **VPAT 2.4** with **axe-core** in CI. **Performance** targets: **<2s p95** dashboard load, **<500ms** API p99, **99.9%** monthly uptime.

OrbitAvanya's team operates as an **embedded Agile squad** within HDR's program structure: **Product Owner (OrbitAvanya)** aligns backlog with HDR's **Technical Lead**; **2 Full-Stack Engineers** deliver frontend/backend; **1 Data Engineer** owns pipelines and models; **1 DevSecOps Engineer** maintains infrastructure, security, and ATO artifacts. **Ceremonies** sync with HDR's **PI Planning** cadence. **IP Rights**: OrbitAvanya retains ownership of the **core platform IP**; HDR receives **unlimited government-purpose rights** to configurations, data, and derivative works created under the subcontract — consistent with **FAR 52.227-14** and **DFARS 252.227-7013**.

This solution eliminates **18+ months of build time**, reduces **TCO by 40–60%** vs. custom development, and positions HDR to offer **analytics-as-a-service** across its **$4.98M+ DOT portfolio** and future task orders.

Core Capabilities

Capability	Description
AI Analytics Dashboard (Core Product)	Production-grade analytics platform with ETL fabric, ML model zoo, and role-based dashboards. Deployed in 3 state DOT / water utility environments. FedRAMP Moderate aligned. Configurable via YAML/JSON — no code changes for new data

	sources or KPIs.
Python ETL/ELT Framework	200+ pre-built connectors (Primavera, ArcGIS, Maximo, SAP, Oracle, CSV, IoT). Incremental CDC, data quality rules (Great Expectations), lineage (OpenLineage). Runs on AWS MWAA / Airflow.
MLOps & Model Governance	End-to-end model lifecycle: training (SageMaker), tracking (MLflow), deployment (FastAPI + ECS), monitoring (Evidently), retraining triggers. Model cards, bias assessments, and explainability (SHAP) built-in.
Geospatial Analytics Engine	PostGIS + Kepler.gl + H3 indexing for asset mapping, corridor analysis, transit accessibility. Supports GTFS/GTFS-RT, GeoJSON, Shapefile. Integrated with dashboard map widgets.
DevSecOps & ATO Acceleration	Terraform IaC for AWS GovCloud landing zone. Checkov/Prowler policy-as-code. ArgoCD GitOps. Continuous monitoring dashboard. SSP, POA&M, SAR templates pre-populated for FedRAMP Moderate.
Section 508 / WCAG 2.1 AA Compliance	Automated axe-core in CI, manual testing by IAAP-certified testers. VPAT 2.4 published per release. Keyboard navigation, screen reader (NVDA/JAWS), color contrast, focus management.

Technology Stack

- React
- Next.js
- Python
- FastAPI
- Node.js
- PostgreSQL
- TimescaleDB
- MongoDB
- AWS (GovCloud)
- ECS Fargate
- MWAA (Airflow)
- SageMaker
- MLflow
- Terraform
- ArgoCD
- GitHub Enterprise
- Apache Kafka
- PostGIS
- Kepler.gl
- dbt
- Great Expectations
- Evidently AI

- LangChain
- Bedrock
- Cognito
- Azure AD Federation

Implementation Timeline

OrbitAvanya proposes an **18-week delivery timeline** structured in **6 phases** aligned with **2-week Agile sprints**. The plan front-loads architecture, security, and data integration (Sprints 0–3) to de-risk ATO and integration dependencies. MVP delivers measurable value by **Week 6**. Production cutover at **Week 9** enables early operational use. Sustainment sprints (Weeks 10–18) add advanced analytics, additional data sources, and knowledge transfer. All phases include **Definition of Done** with security scan, accessibility test, performance benchmark, and HDR Product Owner acceptance.

Engagement Timeline

Phase	Duration	Focus Area	Deliverables
Phase 1: Sprint 0 — Foundation & Security	Week 1–2	AWS Landing Zone, Security Plan (SSP), FedRAMP Control Mapping, Architecture Decision Records, CI/CD Pipeline, GitOps Bootstrap	Deployed AWS GovCloud Landing Zone (Control Tower), SSP v0.1, Control Matrix, Terraform State, ArgoCD Apps, GitHub Repos, IAM Roles, VPC Peering Design
Phase 2: Sprint 1–2 — Data Fabric & Core Ingestion	Week 3–6	ETL Pipeline Development for 5 Priority Sources (Primavera, ArcGIS, Maximo, SAP, IoT), Data Warehouse Schema, dbt Models, Data Quality Gates	5 Production ETL DAGs, PostgreSQL/TimescaleDB Schema, dbt Project, Great Expectations Suites, OpenLineage Integration, Ingestion Runbooks
Phase 3: Sprint 3–4 — MVP Dashboard & Models	Week 7–8	Configure Dashboard (8 Role-Based Views), Deploy 3 Predictive Models (Schedule Risk, Cost Variance, Asset Health), API Layer, RBAC, Section 508 Baseline	MVP Dashboard (Staging), Model Endpoints (FastAPI), MLflow Registry, VPAT 2.4, API Docs (OpenAPI), UAT Test Plan
Phase 4: Sprint 5 — UAT & Production Hardening	Week 9	User Acceptance Testing, Defect Remediation, Performance Tuning, Security Scan (Checkov, Trivy, OWASP ZAP), ATO Package Finalization	UAT Sign-off, Zero Sev 1/2 Defects, Performance Report (<2s p95), SSP v1.0, POA&M, SAR, Continuous Monitoring Plan, Go-Live Runbook
Phase 5: Sprint 6 — Production Cutover & Transition	Week 10	Production Deployment (Blue/Green), DNS Cutover, Monitoring Activation, HDR Operations Handoff, Hypercare Support	Production Dashboard, CloudWatch/Grafana Dashboards, PagerDuty Alerts, Operations Runbook, HDR Admin Training (Session 1)
Phase 6: Sprints 7–10 — Sustainment & Enhancement	Week 11–18	Model Retraining Pipeline, 2 Additional Data Sources, Geospatial Analytics Module, Executive Mobile PWA, Knowledge Transfer Workshops (4)	SageMaker Retraining Pipeline, 2 New ETL DAGs, Kepler.gl Map Widget, Mobile PWA (Capacitor), Admin Guide, Data Steward Playbook, Model Governance Framework, 4 Recorded Workshops

Total Estimated Duration: 18 Weeks

Your Investment

OrbitAvanya proposes a **hybrid Fixed-Price / Time & Materials** pricing structure aligned with delivery milestones and sustained operations. **Fixed-Price** applies to **Phases 1–5 (Weeks 1–10)** covering product configuration, integration, MVP, UAT, and production cutover — providing HDR cost certainty for the core deployment. **T&M** applies to **Phase 6 (Weeks 11–18)** sustainment and enhancement sprints, where scope evolves based on user feedback and emerging task order needs. T&M rates are **ceiling-capped** at **$225,000** with **monthly not-to-exceed (NTE)** guardrails. All pricing assumes **OrbitAvanya-furnished AWS GovCloud infrastructure** (included in fixed price). **Travel/ODC** are **at cost, no fee**, estimated at **$12,000** for 4 on-site workshops. **Payment Terms**: Net 30 upon milestone acceptance. **Invoicing**: Monthly for T&M; milestone-based for Fixed-Price. **Escalation**: 3% annual labor escalation after Year 1 if option years exercised.

Item / Deliverable	Unit	Qty	Unit Price	Total
Phase 1–5 Fixed-Price: Architecture, Security, Data Fabric, MVP, UAT, Production Cutover (Weeks 1–10)	Fixed	1	$485,000	$485,000
Phase 6 T&M Ceiling: Sustainment, Enhancements, Knowledge Transfer (Weeks 11–18)	T&M (Ceiling)	1	$225,000	$225,000
AWS GovCloud Infrastructure (Included in Fixed-Price)	Included	1	$0	$0
Travel & ODC (At Cost, No Fee) — 4 On-Site Workshops	Cost Reimbursable	1	$12,000	$12,000
Option Year 1: Sustainment & Enhancement (12 Months, T&M Ceiling)	T&M (Ceiling)	1	$350,000	$350,000

Service Level Agreement & Terms

- **Availability**: 99.9% monthly uptime (excluding planned maintenance windows: Sat 00:00–04:00 ET)
- **Performance**: Dashboard load <2s p95; API p99 <500ms; ETL SLA: incremental loads <15 min, full refresh <4 hrs
- **Security**: Critical vulnerability (CVSS ≥9.0) patching within **24 hrs**; High (CVSS 7.0–8.9) within **72 hrs**; Monthly **STIG/Config** compliance scan
- **Support Tiers**: **Tier 1 (HDR Help Desk)** — OrbitAvanya provides runbooks, escalation contacts; **Tier 2 (OrbitAvanya)** — 4-hr response (business hours) for Sev 2, 1-hr for Sev 1; **Tier 3 (Product Engineering)** — Root cause within 5 business days
- **Data Integrity**: **RPO <1 hr** (continuous WAL archiving to S3 Cross-Region Replication); **RTO <4 hrs** (validated quarterly via DR drill)
- **Change Management**: All changes via **GitOps PR + ArgoCD**; **Emergency changes** follow **break-glass** process with post-implementation review

- Reporting: Monthly SLA Compliance Report (uptime, latency, error rates, security posture, capacity, backlog health) delivered by 5th business day
- Knowledge Transfer: Quarterly architecture review; Annual DR test participation; On-demand model governance workshop

Company Profile

OrbitAvanya Tech LLP (AvanyaEdge) is a technology services firm specializing in enterprise software development, digital transformation, e-governance platforms, and AI/ML solutions. Founded with a mission to bridge the gap between innovation and government/enterprise delivery, OrbitAvanya brings deep domain expertise across cloud-native architectures, data platforms, cybersecurity, and modern web/mobile application development. Our team of experienced engineers and consultants has successfully delivered multiple government-facing platforms across India and the United States.

Profile Detail	Value
Company Name	OrbitAvanya Tech LLP
Type	Limited Liability Partnership (LLP)
Headquarters	Frisco, Texas 75035, USA
India Office	Pune, Maharashtra, India
Website	www.orbitavanyatech.com
Phone	+917021950643
NAICS Codes	541511, 541512, 541519, 541611
Certifications	MSME (India) · SAM.gov Registered

Appendix

Appendix A — Past Performance

Project	Client	Period	Relevance
State DOT Intelligent Transportation Systems (ITS) Analytics Platform	State Department of Transportation (Southeast Region)	Jan 2022 – Dec 2024	Delivered **AI Analytics Dashboard** for **$12M ITS program** integrating 12 data sources (INRIX, HERE, Wavetronix, RWIS, 511, CAD, SunGuide). 15 predictive models for incident duration, queue formation, travel time reliability. 200+ users. **FedRAMP Moderate ATO achieved**. Reduced manual reporting by **85%**.
Water Utility Asset Performance & Capital Planning Dashboard	Major Metropolitan Water Authority	Mar 2021 – Sep 2023	Configured **AI Analytics Dashboard** for **500+ vertical assets** (pump stations, tanks, treatment plants). Integrated **Maximo, SCADA (OSIsoft PI), GIS, CCTV, CIP Planning**. **Weibull/ML hybrid** deterioration models. **$4.2M** contract. **Section 508 compliant**. Enabled **$18M/yr** capital optimization.
Federal Civilian Agency Enterprise Data Lake & Analytics	U.S. Federal Agency (Cabinet Level)	Oct 2020 – Jun 2022	Built **Python ETL framework** on **AWS GovCloud** for **50+ source systems**. **Airflow + dbt + Great Expectations**. **NIST 800-53 Rev. 5** control implementation (AC, AU, SC, SI families). **CMMI Level 3** processes. Supported **FITARA Category 1** reporting. Team of 8.

Appendix B — Team & Leadership

OrbitAvanya Tech LLP's leadership team comprises professionals with extensive backgrounds in enterprise software delivery, cloud architecture, AI/ML systems, and government program management. Full team profiles and resumes are available upon request.